

# Jiayi Li

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## Education

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| <b>Brown University</b> - Providence, RI                                                 | Exp: May 2026    |
| <i>Master of Science in Computer Science</i>                                             | <b>GPA: 3.75</b> |
| • <b>Coursework:</b> Interactive Computer Graphics, Computational Design and Fabrication |                  |
| <b>Boston University</b> - Boston, MA                                                    | Jan 2024         |
| <i>Bachelor of Arts in Computer Science and Statistics</i>                               | <b>GPA: 3.78</b> |

## Research Experience

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| <b>Research Assistant</b> , Brown University (PI: Shishi Xiao) – Providence, RI                                                                                                                                                           | Jun 2025 – Present |
| • Built an automated data generation pipeline by integrating 4+ AI models (ChatGPT, Flux, SAM, background-removal) in Python, enabling scalable image and text processing for infographic datasets.                                       |                    |
| • Curated a high-quality training dataset by evaluating images image-by-image using defined quality criteria, reducing the pool from 37,000+ raw samples to 11,500+ reliable entries and, improving model consistency.                    |                    |
| • Led monthly research syncs and collaborated on dataset design, creating structured infographic variations through text-to-image prompting strategies and reusable prompt templates.                                                     |                    |
| • Designed an evaluation framework for infographic generation, defining metrics for visual coherence, structural accuracy, and semantic clarity, and benchmarked Style-Aligned to identify performance gaps and guide future comparisons. |                    |

## Professional Experience

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| <b>UI/UX Design Intern</b> , PCCW Global – Beijing, China                                                                                                                                                   | Mar 2024 – Jul 2024 |
| • Led user and competitive research using task-flow mapping and feature benchmarking to uncover key usability gaps, directly shaping the direction of two new internal web product features.                |                     |
| • Proposed 10+ design strategies through interaction analysis and UI trade-off evaluation, improving visual clarity, navigation efficiency, and workflow consistency across the platform.                   |                     |
| • Designed and prototyped responsive interface layouts in close collaboration with PMs, engineers, and UX designers, ensuring design–development alignment and enabling a smoother product handoff process. |                     |

## Projects

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| <b>Custom Matryoshka Doll Design &amp; Fabrication System</b> , Brown University                                                                                                                                                                                                                    | Oct – Dec 2025 |
| • Developed an end-to-end nested-doll fabrication system in C++, integrating clay-style 3D sculpting with real-time mesh deformation using barycentric projection and smooth-falloff algorithms, transforming organic concepts into manufacturable 3D-printable outputs.                            |                |
| • Built a C++/Qt Creator shell-splitting and validation pipeline by implementing parametric inner-shell generation, plane-based mesh segmentation, and real-time collision checking with 3-axis translation controls, enabling reliable fabrication of hollow, interlocking nested-doll components. |                |
| <b>Raft Consensus Implementation (Go, Distributed Systems)</b> , Boston University                                                                                                                                                                                                                  | Jan – Apr 2023 |
| • Improved browser-side stability by implementing the Raft consensus protocol in Go, enabling reliable leader election, log replication, and fault-tolerant coordination across distributed nodes.                                                                                                  |                |
| • Analyzed design trade-offs in a distributed request–response system by running failure-injection tests and timing benchmarks, improving overall system stability and reproducibility.                                                                                                             |                |

## Skills

- **Languages:** C++, C, Java, Python, SQL, JavaScript, Go, R, English (fluent), Mandarin (native)
- **Technologies:** Qt Creator, OpenGL, VSCode, Figma, SQLite, Xcode, RStudio